Chapter 16

The Quantized Electromagnetic Field

16.1 Introduction

Students typically encounter *photons* in their first course on modern physics. Blackbody radiation, the photoelectric effect, Compton scattering, and related observations provide some of the earliest evidence that classical physics is incomplete and help motivate the development of quantum mechanics. More direct experiments involving single-photon detection and photon statistics reveal that the electromagnetic field exchanges energy in discrete units and possesses properties that cannot be reproduced by a purely classical wave theory. It is therefore striking that photons largely disappear from the first serious course in quantum mechanics. The theory is developed almost entirely through systems with a fixed number of particles while the electromagnetic field generally remains classical.

When the quantized electromagnetic field is finally introduced, the usual treatment is predominantly mathematical. One decomposes the classical field into normal modes, identifies each mode with a harmonic oscillator, and then "canonically quantizes" those oscillators by imposing commutation relations inherited from ordinary quantum mechanics. Photons subsequently emerge as oscillator excitations. This procedure is powerful and correct, but its logic can obscure the physical origin of the theory. It does not derive quantum electromagnetism from classical electromagnetism alone. Rather, it combines the classical mode structure with quantum hypotheses and then recovers the photon picture at the end.

Here we take the experimental properties of light as our starting point [1]. We hypothesize that the free electromagnetic field possesses bosonic excitations, called photons, whose energies are $\hbar\omega$, and we describe those excitations using the ordinary framework of quantum mechanics. We then require the resulting quantum theory to correspond to classical source-free electromagnetism in the appropriate limit: its modes must carry the classical wavevector and polarization labels, evolve according to Maxwell's equations, and reproduce the classical electromagnetic energy. In this way, the quantization of the electromagnetic field is organized around the physical observations that motivate it, with the necessary quantum assumptions and correspondence hypotheses stated explicitly rather than hidden inside a formal prescription.

16.2 Classical Source-Free Electromagnetism

Throughout these notes, we take $\epsilon_0 = \mu_0 = c = 1$. Consequently, electric and magnetic fields have the same units, the vacuum dispersion relation is $\omega = |\mathbf{k}|$, and the classical energy is

$$H_{\rm cl} = \frac{1}{2}\int_{\mathbb{R}^3} d^3r \, \left(\mathbf{E}^2 + \mathbf{B}^2\right). \tag{16.1}$$

We retain $\hbar$, since it identifies the genuinely quantum statements. In vacuum and in the absence of charges and currents,

$$\nabla \cdot \mathbf{E} = 0, \qquad \nabla \cdot \mathbf{B} = 0, \tag{16.2}$$

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \qquad \nabla \times \mathbf{B} = \partial_t \mathbf{E}. \tag{16.3}$$

These are the source-free Maxwell equations. With suitable initial data and, when relevant, appropriate boundary conditions, they determine the subsequent evolution of the electromagnetic field. We'll assume the fields are smooth and decrease sufficiently rapidly at spatial infinity. These assumptions ensure that the total field energy is finite and justify the Fourier analysis and integrations by parts used below.

16.2.1 The Wave Equation

Take the curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\partial_t(\nabla \times \mathbf{B}) = -\partial_t^2 \mathbf{E}.$$

Using

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E},$$

where the last equality uses $\nabla \cdot \mathbf{E} = 0$, gives

$$(\partial_t^2 - \nabla^2)\mathbf{E} = 0. \tag{16.4}$$

The same calculation with $\mathbf{B}$ gives

$$(\partial_t^2 - \nabla^2)\mathbf{B} = 0. \tag{16.5}$$

The wave equations are necessary consequences of Maxwell's equations. They are not by themselves sufficient: a solution of the vector wave equation must also satisfy the divergence constraints and the first-order curl equations.

16.2.2 Fourier Transform

Because the source-free Maxwell equations are linear, the Fourier transform allows us to decompose the electric field into plane-wave components that evolve independently. The Fourier transform of

the electric field is

$$\tilde{\mathbf{E}}(\mathbf{k},t) = \frac{1}{(2\pi)^{3/2}} \int d^3r \; e^{-i\mathbf{k}\cdot\mathbf{r}} \; \mathbf{E}(\mathbf{r},t) \tag{16.6}$$

and the inverse transform is

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{(2\pi)^{3/2}} \int d^3k \; e^{i\mathbf{k}\cdot\mathbf{r}} \; \tilde{\mathbf{E}}(\mathbf{k},t) \tag{16.7}$$

where we've taken a symmetric normalization convention. Similar expressions can be written down for the magnetic field $\mathbf{B}$ as well. The Fourier transform decomposes the electromagnetic field into plane-wave modes, allowing us to analyze the evolution of each spatial mode separately. Let $\boldsymbol{\phi}_j$ denote the j-th spatial derivative of the electric field, i.e., $\boldsymbol{\phi}_j = \partial_j \mathbf{E}$. The Fourier transform of this quantity can be determined using integration by parts:

$$\begin{aligned}
\tilde{\boldsymbol{\phi}}_j &= \frac{1}{(2\pi)^{3/2}} \int d^3r \; e^{-i\mathbf{k}\cdot\mathbf{r}} \; \partial_j \mathbf{E} \\
&= \frac{1}{(2\pi)^{3/2}} \left[e^{-i\mathbf{k}\cdot\mathbf{r}} \; \mathbf{E} \right]_{-\infty}^{\infty} - \frac{1}{(2\pi)^{3/2}} \int d^3r \; \partial_j e^{-i\mathbf{k}\cdot\mathbf{r}} \; \mathbf{E} \\
&= ik_j \tilde{\mathbf{E}},
\end{aligned}$$

where the boundary term is zero because the field vanishes at infinity.

Hence, the Fourier transform of the j-th spatial derivative of $\mathbf{E}$ is simply the product of ik_j and the Fourier transform of $\mathbf{E}$. In "k-space," the spatial derivative operator becomes a multiplication operator. Fourier transforming Maxwell's equations gives, for every $\mathbf{k}$,

$$\mathbf{k} \cdot \tilde{\mathbf{E}} = 0, \qquad \mathbf{k} \cdot \tilde{\mathbf{B}} = 0, \tag{16.8}$$

$$i\mathbf{k} \times \tilde{\mathbf{E}} = -\partial_t \tilde{\mathbf{B}}, \qquad i\mathbf{k} \times \tilde{\mathbf{B}} = \partial_t \tilde{\mathbf{E}}. \tag{16.9}$$

and Fourier transforming the wave equation yields

$$\partial_t^2 \tilde{\mathbf{E}} + k^2 \tilde{\mathbf{E}} = 0, \tag{16.10}$$

The general complex solution is

$$\tilde{\mathbf{E}}(\mathbf{k},t) = \mathbf{C}(\mathbf{k}) e^{-i\omega_{\mathbf{k}} t} + \mathbf{D}(\mathbf{k}) e^{i\omega_{\mathbf{k}} t}, \tag{16.11}$$

where $\omega_{\mathbf{k}} = k = |\mathbf{k}|$. Remember, we set $c = 1$; in SI units, the dispersion relation is $\omega_{\mathbf{k}} = ck$. Since the physical field is real-valued, it follows from Eq. (16.6) that

$$\tilde{\mathbf{E}}(\mathbf{k},t)^* = \tilde{\mathbf{E}}(-\mathbf{k},t),$$

and hence, $\mathbf{D}(\mathbf{k}) = \mathbf{C}(-\mathbf{k})^*$. Substituting our solution for $\tilde{\mathbf{E}}$ into Eq. (16.7), we find

$$\begin{aligned}\mathbf{E}(\mathbf{r},t) &= \frac{1}{(2\pi)^{3/2}} \int d^3k \, \left[\mathbf{C}(\mathbf{k}) e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k} t)} + \mathbf{C}(\mathbf{k})^* e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k} t)}\right] \\ &= \frac{2}{(2\pi)^{3/2}} \, \mathrm{Re} \int d^3k \, \mathbf{C}(\mathbf{k}) e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k} t)}\end{aligned} \tag{16.12}$$

Thus, every real solution of the wave equation can be expressed as a continuous superposition of plane waves. All independent information about the field is contained in the complex vector amplitude $\mathbf{C}(\mathbf{k})$, including the wave's amplitude, phase, and polarization.

16.2.3 Polarization

The expression $\mathbf{k}\cdot\tilde{\mathbf{E}} = 0$ implies $\tilde{\mathbf{E}}$ lies in a two-dimensional plane orthogonal to $\mathbf{k}$. For each $\mathbf{k} \neq 0$, we define an orthonormal set of real unit vectors $\{\hat{\mathbf{e}}_\mu(\mathbf{k})\}$ where $\mu = 1, 2$ such that

$$\hat{\mathbf{e}}_\mu(\mathbf{k}) \cdot \hat{\mathbf{e}}_{\mu'}(\mathbf{k}) = \delta_{\mu\mu'}, \quad \mathbf{k}\cdot\hat{\mathbf{e}}_\mu(\mathbf{k}) = 0.$$

It follows that every complex amplitude may be decomposed as

$$\mathbf{C}(\mathbf{k}) = \sum_{\mu=1}^{2} C_\mu(\mathbf{k}) \, \hat{\mathbf{e}}_\mu(\mathbf{k})$$

Thus, for each nonzero wavevector $\mathbf{k}$, the electric field has two independent transverse polarization components. The complex scalar amplitudes $C_1(\mathbf{k})$ and $C_2(\mathbf{k})$ completely specify the amplitude, phase, and polarization of the corresponding plane-wave mode. We may therefore label the classical modes of the free electromagnetic field by the pair $(\mathbf{k}, \mu)$, $\mu = 1, 2$ with frequency $\omega_\mathbf{k} = k$. The choice of the two polarization vectors is not unique. Any other orthonormal basis of the plane perpendicular to $\mathbf{k}$ gives an equivalent description. For example, one may use two linear polarizations or two circular polarizations.

16.2.4 Magnetic Field

Maxwell's equations couple the electric and magnetic fields, specifically through Faraday's law and the Ampère-Maxwell law. The fields are not independent of one another. We can determine the magnetic field from the Fourier-transformed Maxwell equations. In particular, consider the Fourier-transformed Ampère-Maxwell law:

$$i\mathbf{k} \times \tilde{\mathbf{B}} = \partial_t \tilde{\mathbf{E}}$$

Taking the cross product of both sides with $\mathbf{k}$ and using a vector identity,

$$i\mathbf{k} \times \left(\mathbf{k} \times \tilde{\mathbf{B}}\right) = i\mathbf{k}\left(\mathbf{k}\cdot\tilde{\mathbf{B}}\right) - ik^2\tilde{\mathbf{B}} = \mathbf{k} \times \partial_t \tilde{\mathbf{E}}$$

Since $\mathbf{k} \cdot \tilde{\mathbf{B}} = 0$, it follows that

$$\tilde{\mathbf{B}} = \frac{i\mathbf{k}}{k^2} \times \partial_t \tilde{\mathbf{E}}$$

Using the solution for $\tilde{\mathbf{E}}$ and the dispersion relation $\omega_{\mathbf{k}} = k$, it follows that

$$\tilde{\mathbf{B}}(\mathbf{k}, t) = \hat{\mathbf{k}} \times \left[\mathbf{C}(\mathbf{k}) e^{-i\omega_{\mathbf{k}} t} + \mathbf{C}(-\mathbf{k})^* e^{i\omega_{\mathbf{k}} t}\right]$$

From the inverse Fourier transform, we find

$$\mathbf{B}(\mathbf{r}, t) = \frac{2}{(2\pi)^{3/2}} \operatorname{Re} \int d^3k \, \hat{\mathbf{k}} \times \mathbf{C}(\mathbf{k}) \, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} \tag{16.13}$$

Thus, once the complex amplitude $\mathbf{C}(\mathbf{k})$ specifying the electric field is known, the magnetic field is completely determined. In particular, each plane-wave contribution satisfies

$$\mathbf{B}_{\mathbf{k}} = \hat{\mathbf{k}} \times \mathbf{E}_{\mathbf{k}}.$$

16.2.5 Field Energy

Using the plane-wave expansions of $\mathbf{E}$ and $\mathbf{B}$, one can show that the total electromagnetic field energy can be expressed as a continuous sum of contributions from the individual modes $(\mathbf{k}, \mu)$. This is a remarkable claim! One might initially expect the energy to couple different plane-wave modes. After all, the electric and magnetic fields are linear superpositions of plane waves, whereas the electromagnetic energy is quadratic in the fields; squaring these sums naturally produces cross terms between different modes. A detailed calculation shows, however, that all such cross terms vanish. Terms involving different wavevectors or polarizations vanish by orthogonality, while the remaining terms coupling $\mathbf{k}$ and $-\mathbf{k}$ cancel between the electric and magnetic contributions. Consequently, the total energy reduces to

$$H_{\mathrm{cl}} = 2 \sum_{\mu=1}^{2} \int d^3k \, |C_\mu(\mathbf{k})|^2. \tag{16.14}$$

Thus, although the fields are constructed by superposing many plane waves, the energy contains no coupling between distinct modes. It is instead a continuous sum of nonnegative contributions, one from each mode $(\mathbf{k}, \mu)$. In this precise sense, the normal modes of the free electromagnetic field are *energetically independent*.

16.3 The Photon Model

No single experiment establishes the entire photon picture; rather, several independent classes of experiments identify a consistent object carrying energy, momentum, polarization, and bosonic exchange symmetry. We take the following facts as experimental input. The free electromagnetic field has excitations called photons. A photon is associated with a wavevector and one of two transverse polarization states. Its energy is proportional to the frequency of the corresponding electromag-

netic mode, and its momentum is proportional to the wavevector. Photons are indistinguishable bosons: an electromagnetic mode may contain any nonnegative integer number of photons, and multiphoton states are symmetric under the exchange of identical photons. These facts do not follow from Maxwell's equations alone. Maxwell's equations determine the classical modes of the field, but observations supply the additional quantum information about the energies, momenta, polarization states, and statistics of their excitations.

In a modern physics course, you may already have encountered the experimental evidence that light exchanges energy in discrete amounts determined by its frequency and that photons carry momentum determined by their wavevector. We will not review that evidence in detail here. Instead, let us focus on two properties that are often discussed less carefully: the polarization of individual photons and the bosonic statistics obeyed by collections of identical photons. When single photons pass through a polarizer, each photon is either transmitted or absorbed. After many trials, the transmission frequencies reproduce the familiar classical polarization law. For each propagation direction, there are two independent transverse polarization states. These may be represented by two linear polarizations or, equivalently, by right- and left-circular polarization.

The Hong–Ou–Mandel experiment provides one of the clearest demonstrations that photons exhibit bosonic statistics. The experiment involves two photons that enter the two input ports of a balanced beam splitter. When the photons are distinguishable, perhaps because they arrive at slightly different times, detections in separate output ports occur at the expected rate. When the photons are made indistinguishable, these coincidence events disappear; the photons leave together through the same output port. This occurs because the two indistinguishable alternatives that could produce a coincidence have quantum amplitudes that cancel. The experiment effectively probes the exchange symmetry of the two-photon state. Its outcome is precisely the behavior expected for identical bosons.

16.3.1 Discretizing the Modes

In unbounded space, the wavevector $\mathbf{k}$ varies continuously. This leads to continuum-normalized states and Dirac delta functions, which can obscure the basic construction. We therefore begin by restricting the mathematical description to a finite cubic region and imposing periodic boundary conditions. This is a mathematical convenience rather than a physical confinement of the field. Once the construction is complete, we will take the limit $L \to \infty$ and recover the continuum description. Consider a cubic region of volume $V = L^3$, and impose periodic boundary conditions on the fields:

$$\mathbf{E}(\mathbf{r} + L\hat{\mathbf{r}}_j, t) = \mathbf{E}(\mathbf{r}, t), \qquad \mathbf{B}(\mathbf{r} + L\hat{\mathbf{r}}_j, t) = \mathbf{B}(\mathbf{r}, t).$$

A plane wave must then satisfy

$$e^{i\mathbf{k}\cdot(\mathbf{r}+L\hat{\mathbf{r}}_j)} = e^{i\mathbf{k}\cdot\mathbf{r}},$$

and hence

$$e^{ik_jL} = 1.$$

The allowed components of the wavevector are therefore

$$k_j = \frac{2\pi m_j}{L}, \qquad m_j \in \mathbb{Z},$$

so that

$$\mathbf{k} = \frac{2\pi}{L}(m_x, m_y, m_z).$$

The allowed wavevectors form a discrete lattice in wavevector space. For every nonzero $\mathbf{k}$, there are two independent transverse polarizations, so the classical modes are labeled by the discrete pairs $(\mathbf{k}, \mu),\ \mu = 1, 2$.

16.3.2 Photon-Number States

We hypothesize that each classical mode $(\mathbf{k}, \mu)$ of the free electromagnetic field corresponds to an independent "quantum mode." Here $\mathbf{k}$ is the wavevector and $\mu = 1, 2$ labels one of the two transverse polarizations. Because photons are bosons, each mode may contain any nonnegative integer number of photons. We denote the state containing n photons in the mode $(\mathbf{k}, \mu)$ by

$$|n_{\mathbf{k}\mu}\rangle, \qquad n_{\mathbf{k}\mu} = 0, 1, 2, \ldots.$$

This notation describes the occupation of a single mode. A general photon-number state of the entire field must specify the occupation number of every mode. We write such a state as

$$\big|\{n_{\mathbf{k}\mu}\}\big\rangle,$$

where the braces denote the complete collection of occupation numbers. The state in which every occupation number vanishes is the vacuum state,

$$|0\rangle = \big|\{n_{\mathbf{k}\mu} = 0\}\big\rangle.$$

A photon-number state is not a state of N individually labeled photons. Because photons are indistinguishable, the physically relevant information is instead the number of photons occupying each mode. The bosonic symmetry of the multiphoton state is therefore incorporated directly into the occupation-number description. From the classical analysis, the mode labeled by $\mathbf{k}$ has frequency

$$\omega_{\mathbf{k}} = |\mathbf{k}|$$

in units with $c = 1$. We now use the experimental input that each photon in this mode carries energy $\hbar\omega_{\mathbf{k}}$. Since the classical modes are energetically independent, we hypothesize that the energies of photons occupying different quantum modes are additive. The photon-number states are therefore

energy eigenstates satisfying

$$H\big|\{n_{\mathbf{k}\mu}\}\big\rangle = \left(\sum_{\mathbf{k},\mu} \hbar\omega_{\mathbf{k}}\, n_{\mathbf{k}\mu}\right) \big|\{n_{\mathbf{k}\mu}\}\big\rangle. \tag{16.15}$$

This equation defines the action of the free-field Hamiltonian on the photon-number basis. In particular, a state containing n photons in a single mode has energy $n\hbar\omega_{\mathbf{k}}$. We have chosen the vacuum energy to be zero. Adding the same constant to every energy eigenvalue would not affect the dynamics of the free field. The zero-point term that appears in the harmonic-oscillator approach is therefore not required at this stage of the photon-first construction.

16.3.3 Creation and Annihilation Operators

For each electromagnetic mode $(\mathbf{k}, \mu)$, define an annihilation operator $a_{\mathbf{k}\mu}$ and a creation operator $a^\dagger_{\mathbf{k}\mu}$. Their actions on the photon-number states of that mode are

$$a_{\mathbf{k}\mu}|n_{\mathbf{k}\mu}\rangle = \sqrt{n_{\mathbf{k}\mu}}\,|n_{\mathbf{k}\mu} - 1\rangle \tag{16.16}$$

and

$$a^\dagger_{\mathbf{k}\mu}|n_{\mathbf{k}\mu}\rangle = \sqrt{n_{\mathbf{k}\mu} + 1}\,|n_{\mathbf{k}\mu} + 1\rangle. \tag{16.17}$$

The annihilation operator removes one photon from the mode, while the creation operator adds one photon. In particular,

$$a_{\mathbf{k}\mu}|0\rangle = 0, \qquad a^\dagger_{\mathbf{k}\mu}|0\rangle = |1_{\mathbf{k}\mu}\rangle.$$

When these operators act on a state of the entire field, they change only the occupation number of the specified mode:

$$a_{\mathbf{k}\mu}\big|\{n_{\mathbf{q}\sigma}\}\big\rangle = \sqrt{n_{\mathbf{k}\mu}}\,\big|\ldots, n_{\mathbf{k}\mu} - 1, \ldots\big\rangle,$$
$$a^\dagger_{\mathbf{k}\mu}\big|\{n_{\mathbf{q}\sigma}\}\big\rangle = \sqrt{n_{\mathbf{k}\mu} + 1}\,\big|\ldots, n_{\mathbf{k}\mu} + 1, \ldots\big\rangle.$$

All other occupation numbers remain unchanged. The square-root factors encode the normalization associated with bosonic occupation. They also imply that

$$a_{\mathbf{k}\mu}a^\dagger_{\mathbf{k}\mu}|n_{\mathbf{k}\mu}\rangle = (n_{\mathbf{k}\mu} + 1)|n_{\mathbf{k}\mu}\rangle,$$

whereas

$$a^\dagger_{\mathbf{k}\mu}a_{\mathbf{k}\mu}|n_{\mathbf{k}\mu}\rangle = n_{\mathbf{k}\mu}|n_{\mathbf{k}\mu}\rangle.$$

Subtracting gives

$$\left[a_{\mathbf{k}\mu}, a^\dagger_{\mathbf{k}\mu}\right]|n_{\mathbf{k}\mu}\rangle = |n_{\mathbf{k}\mu}\rangle.$$

Since this holds for every photon-number state,

$$\left[a_{\mathbf{k}\mu}, a^\dagger_{\mathbf{k}\mu}\right] = 1.$$

Operators belonging to different modes commute because they change independent occupation numbers. The complete bosonic commutation relations are therefore

$$\boxed{\left[a_{\mathbf{k}\mu}, a^\dagger_{\mathbf{k}'\mu'}\right] = \delta_{\mathbf{k}\mathbf{k}'}\delta_{\mu\mu'},} \qquad \boxed{\left[a_{\mathbf{k}\mu}, a_{\mathbf{k}'\mu'}\right] = \left[a^\dagger_{\mathbf{k}\mu}, a^\dagger_{\mathbf{k}'\mu'}\right] = 0.} \tag{16.18}$$

In this photon-first construction, these relations follow from the definitions of the creation and annihilation operators on the bosonic photon-number basis.

Define the number operator for the mode $(\mathbf{k}, \mu)$ by

$$N_{\mathbf{k}\mu} = a^\dagger_{\mathbf{k}\mu} a_{\mathbf{k}\mu}, \tag{16.19}$$

such that

$$N_{\mathbf{k}\mu}\big|\{n_{\mathbf{q}\sigma}\}\big\rangle = n_{\mathbf{k}\mu}\big|\{n_{\mathbf{q}\sigma}\}\big\rangle. \tag{16.20}$$

The Hamiltonian can now be written as

$$\boxed{H = \sum_{\mathbf{k},\mu} \hbar\omega_{\mathbf{k}} a^\dagger_{\mathbf{k}\mu} a_{\mathbf{k}\mu} = \sum_{\mathbf{k},\mu} \hbar\omega_{\mathbf{k}} N_{\mathbf{k}\mu}} \tag{16.21}$$

In the Heisenberg picture of quantum mechanics, the states remain stationary while the operators evolve unitarily over time. Operators must obey the Heisenberg equation of motion. For an operator A that is independent of time,

$$\frac{dA}{dt} = \frac{i}{\hbar}[H, A].$$

Using $[AB, C] = A[B, C] + [A, C]B$ and the commutation relations,

$$\begin{aligned}
\frac{da_{\mathbf{k}\mu}}{dt} &= \frac{i}{\hbar}\left[\hbar\omega a^\dagger_{\mathbf{k}\mu} a_{\mathbf{k}\mu}, a_{\mathbf{k}\mu}\right] \\
&= i\omega_{\mathbf{k}}\left(a^\dagger_{\mathbf{k}\mu}\left[a_{\mathbf{k}\mu}, a_{\mathbf{k}\mu}\right] + \left[a^\dagger_{\mathbf{k}\mu}, a_{\mathbf{k}\mu}\right] a_{\mathbf{k}\mu}\right) \\
&= -i\omega_{\mathbf{k}} a_{\mathbf{k}\mu}
\end{aligned}$$

The solution for the time-evolved Heisenberg annihilation operator is

$$a_{\mathbf{k}\mu}(t) = e^{-i\omega_{\mathbf{k}}t} a_{\mathbf{k}\mu}, \tag{16.22}$$

and a similar calculation reveals

$$a^\dagger_{\mathbf{k}\mu}(t) = e^{i\omega_{\mathbf{k}}t} a^\dagger_{\mathbf{k}\mu}, \tag{16.23}$$

16.3.4 Electric and Magnetic Field Operators

We have constructed the photon-number Hilbert space, introduced creation and annihilation operators for each mode, and obtained their Heisenberg-picture evolution. We now seek operators representing the electric and magnetic fields. The photon Hilbert space and Hamiltonian do not, by themselves, determine these field operators. We must introduce additional correspondence hypotheses connecting the quantum theory to classical electromagnetism.

The quantum fields must be constructed from the same plane-wave modes $(\mathbf{k}, \mu)$ that appear in the classical theory. Because the classical free field depends linearly on its mode amplitudes, the quantum fields are taken to be linear in the corresponding creation and annihilation operators. The electric and magnetic fields represent observable quantities, so their operators must be Hermitian. The field operators satisfy the source-free Maxwell equations as operator equations. It then follows that their expectation values satisfy the classical Maxwell equations in every quantum state. Finally, the electromagnetic energy calculated from the field operators agrees, up to an additive constant, with the photon Hamiltonian constructed previously.

For each nonzero allowed wavevector $\mathbf{k}$, choose two real polarization vectors $\hat{\mathbf{e}}_{\mathbf{k}\mu}$ satisfying

$$\mathbf{k} \cdot \hat{\mathbf{e}}_{\mathbf{k}\mu} = 0, \qquad \hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{\mathbf{k}\mu'} = \delta_{\mu\mu'}.$$

We then propose the following ansatz for the electric-field operator:

$$\mathbf{E}(\mathbf{r}, t) = i \sum_{\mathbf{k},\mu} \mathcal{E}_{\mathbf{k}}\, \hat{\mathbf{e}}_{\mathbf{k}\mu}\, a_{\mathbf{k}\mu}\, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}}t)} + \text{h.c.}, \tag{16.24}$$

where the overall factor of i is conventional and $\mathcal{E}_{\mathbf{k}}$ is an undetermined complex number. The second term h.c. denotes the Hermitian conjugate of the first term, and it ensures that the operator $\mathbf{E}$ is itself Hermitian. We can always absorb the phase of $\mathcal{E}_{\mathbf{k}}$ into the ladder operators, so for now we'll consider $\mathcal{E}_{\mathbf{k}}$ to be real and positive. The transversality of the polarization vectors automatically satisfies $\nabla \cdot \mathbf{E} = 0$.

We can write down a similar expression for the magnetic field and apply Maxwell's equations to determine a relationship between the coefficients. The magnetic field operator can then be written as

$$\mathbf{B}(\mathbf{r}, t) = i \sum_{\mathbf{k},\mu} \mathcal{E}_{\mathbf{k}} \Big(\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\mathbf{k}\mu} \Big) a_{\mathbf{k}\mu} e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}}t)} + \text{h.c.} \tag{16.25}$$

as long as $\omega_{\mathbf{k}} = |\mathbf{k}|$. The field operators therefore satisfy all four source-free Maxwell equations.

Maxwell's equations determine the relationship between the electric and magnetic mode amplitudes, but they do not determine the magnitude of $\mathcal{E}_{\mathbf{k}}$.

We now determine $\mathcal{E}_{\mathbf{k}}$ by requiring that the energy calculated from the fields agrees with the photon Hamiltonian up to an overall constant. In our units,

$$H_{\text{field}} = \frac{1}{2}\int_V d^3r \, \left(\mathbf{E}^2 + \mathbf{B}^2\right).$$

Using the expressions for $\mathbf{E}$ and $\mathbf{B}$, we find

$$H_{\text{field}} = V\sum_{\mathbf{k},\mu} \mathcal{E}_{\mathbf{k}}^2 \left(a_{\mathbf{k}\mu}a_{\mathbf{k}\mu}^\dagger + a_{\mathbf{k}\mu}^\dagger a_{\mathbf{k}\mu}\right). \tag{16.26}$$

The calculation is straightforward but lengthy; see 16.6. The only non-trivial step is to recognize that, with discrete wavevectors,

$$\int_V d^3r \, e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} = V\delta_{\mathbf{k}\mathbf{k}'}.$$

In terms of the number operator, the field energy is

$$H_{\text{field}} = 2V\sum_{\mathbf{k},\mu} \mathcal{E}_{\mathbf{k}}^2 \left(N_{\mathbf{k}\mu} + \frac{1}{2}\right). \tag{16.27}$$

For the occupation-dependent portion to agree with the photon Hamiltonian constructed earlier, we require

$$2V\mathcal{E}_{\mathbf{k}}^2 = \hbar\omega_{\mathbf{k}}$$

and hence,

$$\mathcal{E}_{\mathbf{k}} = \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2V}} \tag{16.28}$$

Substituting this back into the field energy gives

$$\boxed{H_{\text{field}} = \sum_{\mathbf{k},\mu} \hbar\omega_{\mathbf{k}} \left(N_{\mathbf{k}\mu} + \frac{1}{2}\right).} \tag{16.29}$$

The first term is the photon Hamiltonian, while

$$E_{\text{vac}} = \frac{1}{2}\sum_{\mathbf{k},\mu} \hbar\omega_{\mathbf{k}}$$

is the vacuum energy, also called the zero-point energy. This constant has no effect on the free-field dynamics because it commutes with every operator. Consequently, both Hamiltonians generate the same Heisenberg equations and give the same energy differences between photon-number states.

Substituting the expression for $\mathcal{E}_{\mathbf{k}}$ into the electric- and magnetic-field operators gives

$$\boxed{\mathbf{E}(\mathbf{r},t) = i\sum_{\mathbf{k},\mu}\sqrt{\frac{\hbar\omega_k}{2V}}\,\hat{\mathbf{e}}_{\mathbf{k}\mu}a_{\mathbf{k}\mu}e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_k t)} + \text{h.c.}}$$

and

$$\boxed{\mathbf{B}(\mathbf{r},t) = i\sum_{\mathbf{k},\mu}\sqrt{\frac{\hbar\omega_k}{2V}}\left(\hat{\mathbf{k}}\times\hat{\mathbf{e}}_{\mathbf{k}\mu}\right)a_{\mathbf{k}\mu}e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_k t)} + \text{h.c.}}$$

Assuming that our procedure has given us the correct expressions for the electric- and magnetic-field operators, we now turn our attention to the genuinely quantum-mechanical features of the quantized electromagnetic field. In a photon-number state, the expectation values of the electric and magnetic fields vanish. This is because each field operator contains creation and annihilation operators, which map a photon-number state to orthogonal states. The expectation values of the squared fields, however, do not vanish. Thus, a photon-number state can possess nonzero electromagnetic energy and *nonzero field fluctuations* even though its mean electric and magnetic fields are zero. The vacuum provides the simplest example: its mean fields vanish, but its mean-square fields are nonzero.

A nonzero mean electric or magnetic field requires a coherent superposition of states with different photon numbers. More precisely, because the field operators change photon number by one, the state must contain coherence between photon-number states whose occupation numbers differ by one. An incoherent statistical mixture of different photon numbers still has a vanishing mean field. Coherent states are the most important example of states with nonzero mean fields. Their field expectation values have the same plane-wave form as classical electromagnetic fields, while their photon numbers are not definite. When the mean photon number is large, the relative photon-number fluctuations become small, and the coherent state increasingly resembles a classical wave.

Example 16.1: Photon-Number States

Let $|\{n_{\mathbf{k}\mu}\}\rangle$ denote a photon-number state in a finite periodic box. Calculate the expectation values of $\mathbf{E}$, $\mathbf{B}$, $\mathbf{E}^2$, and $\mathbf{B}^2$. Use the last two expressions to calculate the expectation value of H_{field} and confirm that it gives the correct energy.

Solution. For every mode,

$$\langle\{n\}|a_{\mathbf{k}\mu}|\{n\}\rangle = 0, \qquad \langle\{n\}|a^\dagger_{\mathbf{k}\mu}|\{n\}\rangle = 0\rangle\,,$$

because the ladder operators change the photon occupation number. Therefore,

$$\langle\mathbf{E}(\mathbf{r},t)\rangle = 0, \qquad \langle\mathbf{B}(\mathbf{r},t)\rangle = 0.$$

For products of ladder operators,

$$\left\langle a_{\mathbf{k}\mu} a_{\mathbf{k}'\mu'} \right\rangle = \left\langle a^\dagger_{\mathbf{k}\mu} a^\dagger_{\mathbf{k}'\mu'} \right\rangle = 0,$$

while

$$\left\langle a^\dagger_{\mathbf{k}\mu} a_{\mathbf{k}'\mu'} \right\rangle = n_{\mathbf{k}\mu} \delta_{\mathbf{k}\mathbf{k}'} \delta_{\mu\mu'}$$

and

$$\left\langle a_{\mathbf{k}\mu} a^\dagger_{\mathbf{k}'\mu'} \right\rangle = (n_{\mathbf{k}\mu} + 1)\, \delta_{\mathbf{k}\mathbf{k}'} \delta_{\mu\mu'}.$$

Using $\hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{\mathbf{k}\mu} = 1$, we obtain

$$\begin{aligned} \langle \mathbf{E}^2(\mathbf{r}, t) \rangle &= \sum_{\mathbf{k},\mu} \frac{\hbar\omega_{\mathbf{k}}}{2V} \left[\left\langle a_{\mathbf{k}\mu} a^\dagger_{\mathbf{k}\mu} \right\rangle + \left\langle a^\dagger_{\mathbf{k}\mu} a_{\mathbf{k}\mu} \right\rangle \right] \\ &= \sum_{\mathbf{k},\mu} \frac{\hbar\omega_{\mathbf{k}}}{2V} \left[(n_{\mathbf{k}\mu} + 1) + n_{\mathbf{k}\mu} \right]. \end{aligned}$$

Hence,

$$\boxed{\langle \mathbf{E}^2(\mathbf{r}, t) \rangle = \sum_{\mathbf{k},\mu} \frac{\hbar\omega_{\mathbf{k}}}{2V} \left(2n_{\mathbf{k}\mu} + 1 \right).}$$

The magnetic field calculation gives the same result, i.e., $\left\langle \mathbf{E}^2 \right\rangle = \left\langle \mathbf{B}^2 \right\rangle$. The expectation value of the field Hamiltonian is

$$\begin{aligned} \langle H \rangle &= \frac{1}{2} \int_V d^3 r \, \left(\langle \mathbf{E}^2 \rangle + \langle \mathbf{B}^2 \rangle \right) \\ &= \frac{1}{2} V \sum_{\mathbf{k},\mu} \frac{\hbar\omega_{\mathbf{k}}}{V} \left(2n_{\mathbf{k}\mu} + 1 \right) \\ &= \sum_{\mathbf{k},\mu} \hbar\omega_{\mathbf{k}} \left(n_{\mathbf{k}\mu} + \frac{1}{2} \right). \end{aligned}$$

This agrees with the result obtained from computing the expectation value directly using $\langle \{n\} | H | \{n\} \rangle$; it's just the energy eigenvalue of the photon-number state.

Hence, for an electromagnetic field prepared in a photon-number state,

$$\langle \mathbf{E} \rangle = 0, \qquad \langle \mathbf{E}^2 \rangle > 0.$$

The first equation says that the probability distribution for a suitable electric-field measurement is centered at zero. The second says that the distribution has nonzero width. Therefore, individual measurements need not return zero; repeated measurements on identically pre-

pared states produce different outcomes whose average approaches zero.

16.4 The Jaynes–Cummings Model

Consider a generic two-level system with ground state $|g\rangle$, excited state $|e\rangle$, and energy separation $\hbar\omega_0$. Its free Hamiltonian may be written as

$$H_{\text{atom}} = \frac{\hbar\omega_0}{2}\sigma_z,$$

where

$$\sigma_z = |e\rangle\langle e| - |g\rangle\langle g|.$$

Define the raising and lowering operators

$$\sigma_+ = |e\rangle\langle g|, \qquad \sigma_- = |g\rangle\langle e|.$$

Suppose that the two-level system interacts primarily with one electromagnetic mode of frequency ω, wavevector $\mathbf{k}$, and polarization μ. The free-field Hamiltonian for this mode is

$$H_{\text{field}} = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right).$$

The free Hamiltonian of the combined system is therefore

$$H_0 = \frac{\hbar\omega_0}{2}\sigma_z + \hbar\omega\left(a^\dagger a + \frac{1}{2}\right).$$

Assuming the wavelength of the radiation is much larger than the characteristic size of the two-level system, we use the *electric-dipole approximation* to model the interaction Hamiltonian as

$$V = -\mathbf{d}\cdot\mathbf{E}(\mathbf{r}_0)$$

where $\mathbf{d}$ is the dipole operator of the two-level system and $\mathbf{r}_0$ is the location of the two-level system. Many two-level systems lack a permanent dipole moment in their energy eigenstates, so within the two-level subspace,

$$\mathbf{d} = \mathbf{d}_{eg}\sigma_+ + \mathbf{d}_{ge}\sigma_-,$$

where

$$\mathbf{d}_{eg} = \langle e|\mathbf{d}|g\rangle, \quad \mathbf{d}_{ge} = \mathbf{d}_{eg}^*.$$

For the selected mode, the Schrödinger-picture electric-field operator at the position of the two-level system is

$$\mathbf{E}(\mathbf{r}_0) = i\mathcal{E}\,\hat{\mathbf{e}}\left(ae^{i\mathbf{k}\cdot\mathbf{r}_0} - a^\dagger e^{-i\mathbf{k}\cdot\mathbf{r}_0}\right) = \mathcal{E}\,\hat{\mathbf{e}}\left(a + a^\dagger\right),$$

where $\mathcal{E} = \sqrt{\hbar\omega/2V}$, and in the second equality, we've absorbed the constant phases coming from i and $e^{i\mathbf{k}\cdot\mathbf{r}_0}$ into the definitions of the ladder operators. In other words,

$$iae^{i\mathbf{k}\cdot\mathbf{r}_0} \longrightarrow a$$

The interaction Hamiltonian is

$$\begin{aligned} V &= -\left(\mathbf{d}_{eg}\sigma_+ + \mathbf{d}_{ge}\sigma_-\right)\cdot\left[\mathcal{E}\widehat{\mathbf{e}}(a+a^\dagger)\right] \\ &= -\mathcal{E}\left[\left(\mathbf{d}_{eg}\cdot\widehat{\mathbf{e}}\right)\sigma_+ + \left(\mathbf{d}_{ge}\cdot\widehat{\mathbf{e}}\right)\sigma_-\right](a+a^\dagger). \end{aligned}$$

The atomic and field operators act on different Hilbert spaces, so they commute. We may expand the product:

$$\begin{aligned} V = -\mathcal{E}\Big[&\left(\mathbf{d}_{eg}\cdot\widehat{\mathbf{e}}\right)a\sigma_+ + \left(\mathbf{d}_{eg}\cdot\widehat{\mathbf{e}}\right)a^\dagger\sigma_+ \\ &+ \left(\mathbf{d}_{ge}\cdot\widehat{\mathbf{e}}\right)a\sigma_- + \left(\mathbf{d}_{ge}\cdot\widehat{\mathbf{e}}\right)a^\dagger\sigma_-\Big]. \end{aligned}$$

For a real polarization vector,

$$\mathbf{d}_{ge}\cdot\widehat{\mathbf{e}} = \left(\mathbf{d}_{eg}\cdot\widehat{\mathbf{e}}\right)^*.$$

Define the generally complex coupling constant

$$\hbar g = -\mathcal{E}\left(\mathbf{d}_{eg}\cdot\widehat{\mathbf{e}}\right).$$

Then

$$\hbar g^* = -\mathcal{E}\left(\mathbf{d}_{ge}\cdot\widehat{\mathbf{e}}\right),$$

and the interaction becomes

$$V = \hbar g\left(a\sigma_+ + a^\dagger\sigma_+\right) + \hbar g^*\left(a\sigma_- + a^\dagger\sigma_-\right).$$

The phase of g can be absorbed into the relative phases of $|e\rangle$ and $|g\rangle$. We may therefore choose g to be real and positive. Its magnitude is

$$g = \frac{\mathcal{E}}{\hbar}\left|\mathbf{d}_{eg}\cdot\widehat{\mathbf{e}}\right| = \left|\mathbf{d}_{eg}\cdot\widehat{\mathbf{e}}\right|\sqrt{\frac{\omega}{2\hbar V}}.$$

After making this phase choice,

$$V = \hbar g\left(a\sigma_+ + a^\dagger\sigma_+ + a\sigma_- + a^\dagger\sigma_-\right)$$

The term $a\sigma_+$ excites the two-level system while annihilating a photon. It therefore describes *photon absorption*. The term $a^\dagger\sigma_-$ lowers the two-level system while creating a photon, thus describing *photon emission*. The free-energy difference between states involved in photon absorption or emission is $\hbar|\omega - \omega_0|$, which is small when the field is nearly resonant with the transition. The

remaining terms simultaneously excite both systems or simultaneously lower both systems. These terms couple states whose free energies differ by $\hbar(\omega+\omega_0)$. Consequently, the relative phases of the coupled states vary rapidly at frequency $\omega+\omega_0$, causing the effects of these terms to largely average out over the longer time scale of the near-resonant dynamics. When the coupling is weak compared with $\omega+\omega_0$, we therefore neglect these two terms. This is the *rotating-wave approximation*, and we write

$$V \approx V_{\mathrm{RWA}} = \hbar g \left(a\sigma_+ + a^\dagger \sigma_- \right)$$

The rotating wave approximation is valid in the near-resonant, weak-coupling regime.

With the rotating-wave approximation, the Hamiltonian describing the two-level system coupled to a single mode of the electromagnetic field is

$$\boxed{H = \frac{\hbar\omega_0}{2}\sigma_z + \hbar\omega\left(a^\dagger a + \frac{1}{2}\right) + \hbar g\left(a\sigma_+ + a^\dagger\sigma_-\right)} \tag{16.30}$$

This is known as the *Jaynes–Cummings Hamiltonian*, and it is the simplest model in which both matter and the electromagnetic field are treated quantum mechanically. The interaction term describes the coherent exchange of an excitation between the two-level system and the field. Despite its simplicity, the Jaynes–Cummings model captures many of the central features of quantum light–matter interaction, including photon absorption and emission, vacuum Rabi oscillations, stimulated emission, atom–field entanglement, and the preparation and transfer of single-photon states. It applies approximately to systems such as atoms in optical or microwave cavities, Rydberg atoms, quantum dots, and superconducting qubits coupled to resonators. Its validity relies on the single-mode, two-level, electric-dipole, and rotating-wave approximations, together with the neglect of dissipation.

In the Jaynes–Cummings model, the total number of excitations is conserved. For example, the state $|e,n\rangle$ describes an excited two-level system and n photons while $|g,n+1\rangle$ contains $n+1$ photons and a ground-state two-level system. Both states have the same total excitation number $n+1$, and the interaction couples them in the sense that

$$a^\dagger\sigma_- |e,n\rangle = \sqrt{n+1}\,|g,n+1\rangle\,, \quad a\sigma_+ |g,n+1\rangle = \sqrt{n+1}\,|e,n\rangle\,.$$

Thus, for every n, the dynamics is confined to the two-dimensional subspace

$$\mathrm{span}\,\{|e,n\rangle,\ |g,n+1\rangle\}\,.$$

Another way to understand this result is to define the total excitation number operator

$$N_{\mathrm{tot}} = a^\dagger a + |e\rangle\langle e|$$

and then show that $[H, N_{\mathrm{tot}}] = 0$. Thus, the total excitation number is conserved. This conservation law divides the Hilbert space into independent subspaces. Evidently, despite the nonzero vacuum

fluctuations of the electromagnetic field, the state $|g,0\rangle$ cannot become excited in the Jaynes–Cummings model; the two-level system is already in its ground state, and the field contains no excitation that it can absorb.

The model does, however, describe vacuum-induced emission into the cavity mode, a process that cannot be explained by a semiclassical model with a vanishing classical field. An initially excited two-level system in the state $|e,0\rangle$ is coupled to $|g,1\rangle$, so it can decay to its ground state while creating a photon in the initially empty mode. In the ideal single-mode Jaynes–Cummings model, this process is reversible and produces oscillations rather than irreversible free-space decay.

Example 16.2: Rabi Oscillations

Consider a two-level system resonantly coupled to a single mode of the electromagnetic field, so that $\omega = \omega_0$. Initially, the two-level system is in its excited state and the cavity mode is in the vacuum state:

$$|\psi(0)\rangle = |e,0\rangle\,.$$

Determine the state $|\psi(t)\rangle$ at time t. Then find the probability that the system has emitted a photon into the cavity and the earliest time at which this probability reaches its maximum value.

Solution. The Jaynes–Cummings Hamiltonian conserves the total number of excitations. The initial state $|e,0\rangle$ contains one excitation, so the evolution remains in the one-excitation subspace. Hence,

$$|\psi(t)\rangle = c_e(t)\,|e,0\rangle + c_g(t)\,|g,1\rangle\,.$$

The coefficient $c_e(t)$ is the amplitude for the two-level system to remain excited with no photon in the cavity. The coefficient $c_g(t)$ is the amplitude for the two-level system to be in its ground state with one photon in the cavity.

In the ordered basis $\{|e,0\rangle\,, |g,1\rangle\}$, the Hamiltonian is

$$H = \begin{pmatrix} \langle e,0|H|e,0\rangle & \langle e,0|H|g,1\rangle \\ \langle g,1|H|e,0\rangle & \langle g,1|H|g,1\rangle \end{pmatrix} = \hbar \begin{pmatrix} \omega & g \\ g & \omega \end{pmatrix}.$$

The eigenvectors are easy to find; denote them by

$$|+\rangle = \frac{1}{\sqrt{2}}\left(|e,0\rangle + |g,1\rangle\right)$$

and

$$|-\rangle = \frac{1}{\sqrt{2}}\left(|e,0\rangle - |g,1\rangle\right)$$

The corresponding energies are $E_\pm = \hbar(\omega \pm g)$. The time-evolution operator is

$$U(t) = e^{-iHt/\hbar} = \frac{e^{-i(\omega+g)t}}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{e^{-i(\omega-g)t}}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

We separate the common factor of $e^{-i\omega t}$ and simplify using Euler's identity:

$$U(t) = e^{-i\omega t}\begin{pmatrix} \cos(gt) & -i\sin(gt) \\ -i\sin(gt) & \cos(gt) \end{pmatrix}.$$

The complete time-evolved state is

$$|\psi(t)\rangle = e^{-i\omega t}\Big[\cos(gt)\,|e,0\rangle - i\sin(gt)\,|g,1\rangle\Big]$$

and the probability the system has emitted a photon into the cavity is

$$P_{\text{emit}}(t) = |c_g(t)|^2 = \sin^2(gt)$$

The probability is first maximized when $gt = \pi/2$, or

$$t_{\max} = \frac{\pi}{2g}$$

At this time, $\psi(t_{\max}) = |g,1\rangle$. The excitation has been transferred completely from the two-level system to the cavity field. Because the ideal cavity is lossless, the photon will subsequently be reabsorbed, and the excitation will continue to oscillate between $|e,0\rangle$ and $|g,1\rangle$.

16.5 Continuum Limit

We now let $V = L^3 \to \infty$. Since the spacing between neighboring components of the wavevector is $\Delta k_j = 2\pi/L$, one allowed wavevector occupies a volume $\Delta^3 k = (2\pi)^3/V$ in wavevector space. In the continuum limit, we replace sums with integrals:

$$\sum_{\mathbf{k}} \longrightarrow \frac{V}{(2\pi)^3}\int d^3k,$$

and we rescale the ladder operators by introducing

$$a_\mu(\mathbf{k}) = \sqrt{\frac{V}{(2\pi)^3}}\,a_{\mathbf{k}\mu} \tag{16.31}$$

In the continuum limit, the discrete commutation relations are replaced by

$$\begin{aligned}\left[a_\mu(\mathbf{k}), a^\dagger_{\mu'}(\mathbf{k}')\right] &= \delta_{\mu\mu'}\delta^{(3)}(\mathbf{k}-\mathbf{k}'),\\ \left[a_\mu(\mathbf{k}), a_{\mu'}(\mathbf{k}')\right] &= \left[a^\dagger_\mu(\mathbf{k}), a^\dagger_{\mu'}(\mathbf{k}')\right] = 0.\end{aligned} \tag{16.32}$$

The continuum electric-field operator is then

$$\boxed{\mathbf{E}(\mathbf{r},t) = \frac{i}{(2\pi)^{3/2}}\sum_{\mu=1}^{2}\int d^3k\,\sqrt{\frac{\hbar\omega_\mathbf{k}}{2}}\,\hat{\mathbf{e}}_{\mathbf{k}\mu}a_\mu(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{h.c.}} \tag{16.33}$$

Similarly, the magnetic-field operator becomes

$$\boxed{\mathbf{B}(\mathbf{r},t) = \frac{i}{(2\pi)^{3/2}}\sum_{\mu=1}^{2}\int d^3k\,\sqrt{\frac{\hbar\omega_\mathbf{k}}{2}}\,\left(\hat{\mathbf{k}}\times\hat{\mathbf{e}}_{\mathbf{k}\mu}\right)a_\mu(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{h.c.}} \tag{16.34}$$

Lastly, the continuum field Hamiltonian is

$$\boxed{H_\text{field} = \sum_{\mu=1}^{2}\int d^3k\,\hbar\omega_\mathbf{k}a^\dagger_\mu(\mathbf{k})a_\mu(\mathbf{k}) + \frac{V}{2}\sum_{\mu=1}^{2}\int\frac{d^3k}{(2\pi)^3}\,\hbar\omega_\mathbf{k}.} \tag{16.35}$$

Notice that the vacuum energy becomes infinite for two conceptually distinct reasons. First, the total vacuum energy is proportional to the volume V, as expected for a uniform energy density, and therefore becomes infinite when the spatial volume is taken to infinity. Second, the vacuum-energy density is itself divergent because the integral includes modes with arbitrarily large wavevectors. This latter divergence is called an *ultraviolet divergence*.

16.6 Field Energy Calculation

Assume that the phases of $\mathcal{E}_\mathbf{k}$ have been absorbed into the ladder operators, so that $\mathcal{E}_\mathbf{k}$ is real and nonnegative. We also take the polarization vectors to be real. Starting from

$$\mathbf{E}(\mathbf{r},t) = i\sum_{\mathbf{k},\mu}\hat{\mathbf{e}}_{\mathbf{k}\mu}\mathcal{E}_\mathbf{k}a_{\mathbf{k}\mu}e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{h.c.},$$

the Hermitian-conjugate term is

$$\left[i\hat{\mathbf{e}}_{\mathbf{k}\mu}\mathcal{E}_\mathbf{k}a_{\mathbf{k}\mu}e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)}\right]^\dagger = -i\hat{\mathbf{e}}_{\mathbf{k}\mu}\mathcal{E}_\mathbf{k}a^\dagger_{\mathbf{k}\mu}e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)}.$$

Thus,

$$\mathbf{E}(\mathbf{r},t) = i\sum_{\mathbf{k},\mu}\mathcal{E}_\mathbf{k}\hat{\mathbf{e}}_{\mathbf{k}\mu}\left[a_{\mathbf{k}\mu}e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} - a^\dagger_{\mathbf{k}\mu}e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)}\right].$$

Similarly,

$$\mathbf{B}(\mathbf{r},t) = i\sum_{\mathbf{k},\mu} \mathcal{E}_{\mathbf{k}} \left(\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\mathbf{k}\mu}\right) \left[a_{\mathbf{k}\mu} e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_{\mathbf{k}}t)} - a^{\dagger}_{\mathbf{k}\mu} e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega_{\mathbf{k}}t)}\right].$$

Define

$$\theta_{\mathbf{k}} = \mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t$$

and introduce the magnetic polarization vector

$$\hat{\mathbf{b}}_{\mathbf{k}\mu} = \hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\mathbf{k}\mu}.$$

The fields can then be written compactly as

$$\mathbf{E} = i\sum_{\mathbf{k},\mu} \mathcal{E}_{\mathbf{k}} \hat{\mathbf{e}}_{\mathbf{k}\mu} \left(a_{\mathbf{k}\mu} e^{i\theta_{\mathbf{k}}} - a^{\dagger}_{\mathbf{k}\mu} e^{-i\theta_{\mathbf{k}}}\right)$$

and

$$\mathbf{B} = i\sum_{\mathbf{k},\mu} \mathcal{E}_{\mathbf{k}} \hat{\mathbf{b}}_{\mathbf{k}\mu} \left(a_{\mathbf{k}\mu} e^{i\theta_{\mathbf{k}}} - a^{\dagger}_{\mathbf{k}\mu} e^{-i\theta_{\mathbf{k}}}\right).$$

We now substitute these expressions into

$$H_{\text{field}} = \frac{1}{2}\int_V d^3r \, \left(\mathbf{E}^2 + \mathbf{B}^2\right).$$

Electric-Field Contribution

Squaring the electric field gives

$$\begin{aligned}\mathbf{E}^2 = & -\sum_{\mathbf{k},\mu}\sum_{\mathbf{k}',\mu'} \mathcal{E}_{\mathbf{k}}\mathcal{E}_{\mathbf{k}'} \left(\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{e}}_{\mathbf{k}'\mu'}\right) \\ & \times \left(a_{\mathbf{k}\mu} e^{i\theta_{\mathbf{k}}} - a^{\dagger}_{\mathbf{k}\mu} e^{-i\theta_{\mathbf{k}}}\right)\left(a_{\mathbf{k}'\mu'} e^{i\theta_{\mathbf{k}'}} - a^{\dagger}_{\mathbf{k}'\mu'} e^{-i\theta_{\mathbf{k}'}}\right),\end{aligned}$$

where the overall minus sign comes from $i^2 = -1$. Expanding the product,

$$\begin{aligned}\mathbf{E}^2 = \sum_{\mathbf{k},\mu}\sum_{\mathbf{k}',\mu'} \mathcal{E}_{\mathbf{k}}\mathcal{E}_{\mathbf{k}'} \left(\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{e}}_{\mathbf{k}'\mu'}\right) \Big[& - a_{\mathbf{k}\mu} a_{\mathbf{k}'\mu'} e^{i(\theta_{\mathbf{k}}+\theta_{\mathbf{k}'})} \\ & + a_{\mathbf{k}\mu} a^{\dagger}_{\mathbf{k}'\mu'} e^{i(\theta_{\mathbf{k}}-\theta_{\mathbf{k}'})} \\ & + a^{\dagger}_{\mathbf{k}\mu} a_{\mathbf{k}'\mu'} e^{-i(\theta_{\mathbf{k}}-\theta_{\mathbf{k}'})} \\ & - a^{\dagger}_{\mathbf{k}\mu} a^{\dagger}_{\mathbf{k}'\mu'} e^{-i(\theta_{\mathbf{k}}+\theta_{\mathbf{k}'})}\Big].\end{aligned}$$

Magnetic-Field Contribution

In exactly the same way,

$$\begin{aligned}\mathbf{B}^2 = \sum_{\mathbf{k},\mu}\sum_{\mathbf{k}',\mu'} \mathcal{E}_{\mathbf{k}}\mathcal{E}_{\mathbf{k}'}\left(\hat{\mathbf{b}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{b}}_{\mathbf{k}'\mu'}\right)\Big[&- a_{\mathbf{k}\mu}a_{\mathbf{k}'\mu'}e^{i(\theta_{\mathbf{k}}+\theta_{\mathbf{k}'})} \\ &+ a_{\mathbf{k}\mu}a^{\dagger}_{\mathbf{k}'\mu'}e^{i(\theta_{\mathbf{k}}-\theta_{\mathbf{k}'})} \\ &+ a^{\dagger}_{\mathbf{k}\mu}a_{\mathbf{k}'\mu'}e^{-i(\theta_{\mathbf{k}}-\theta_{\mathbf{k}'})} \\ &- a^{\dagger}_{\mathbf{k}\mu}a^{\dagger}_{\mathbf{k}'\mu'}e^{-i(\theta_{\mathbf{k}}+\theta_{\mathbf{k}'})}\Big].\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbf{E}^2 + \mathbf{B}^2 = \sum_{\mathbf{k},\mu}\sum_{\mathbf{k}',\mu'} \mathcal{E}_{\mathbf{k}}\mathcal{E}_{\mathbf{k}'}D_{\mathbf{k}\mu,\mathbf{k}'\mu'}\Big[&- a_{\mathbf{k}\mu}a_{\mathbf{k}'\mu'}e^{i(\theta_{\mathbf{k}}+\theta_{\mathbf{k}'})} \\ &+ a_{\mathbf{k}\mu}a^{\dagger}_{\mathbf{k}'\mu'}e^{i(\theta_{\mathbf{k}}-\theta_{\mathbf{k}'})} \\ &+ a^{\dagger}_{\mathbf{k}\mu}a_{\mathbf{k}'\mu'}e^{-i(\theta_{\mathbf{k}}-\theta_{\mathbf{k}'})} \\ &- a^{\dagger}_{\mathbf{k}\mu}a^{\dagger}_{\mathbf{k}'\mu'}e^{-i(\theta_{\mathbf{k}}+\theta_{\mathbf{k}'})}\Big],\end{aligned}$$

where

$$D_{\mathbf{k}\mu,\mathbf{k}'\mu'} = \hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{e}}_{\mathbf{k}'\mu'} + \hat{\mathbf{b}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{b}}_{\mathbf{k}'\mu'}.$$

There are now two different types of spatial integrals.

Terms Containing One Creation and One Annihilation Operator

For these terms, the spatial exponential is

$$e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}.$$

Periodic plane-wave orthogonality gives

$$\int_V d^3r\, e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} = V\delta_{\mathbf{k}\mathbf{k}'}.$$

Thus, only $\mathbf{k}' = \mathbf{k}$ survives. For equal wavevectors,

$$\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{e}}_{\mathbf{k}\mu'} = \delta_{\mu\mu'}.$$

Furthermore,

$$\begin{aligned}\hat{\mathbf{b}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{b}}_{\mathbf{k}\mu'} &= \left(\hat{\mathbf{k}}\times\hat{\mathbf{e}}_{\mathbf{k}\mu}\right)\cdot\left(\hat{\mathbf{k}}\times\hat{\mathbf{e}}_{\mathbf{k}\mu'}\right) \\ &= \left(\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}\right)\left(\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{e}}_{\mathbf{k}\mu'}\right) - \left(\hat{\mathbf{k}}\cdot\hat{\mathbf{e}}_{\mathbf{k}\mu'}\right)\left(\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{k}}\right)\end{aligned}$$

$$= \delta_{\mu\mu'}.$$

Therefore,

$$D_{\mathbf{k}\mu,\mathbf{k}\mu'} = 2\delta_{\mu\mu'}.$$

The spatial integral sets both $\mathbf{k}' = \mathbf{k}$ and $\mu' = \mu$. The time-dependent phase then disappears:

$$e^{i(\theta_{\mathbf{k}} - \theta_{\mathbf{k}})} = 1.$$

These terms contribute

$$2V \sum_{\mathbf{k},\mu} \mathcal{E}_{\mathbf{k}}^2 \left(a_{\mathbf{k}\mu} a_{\mathbf{k}\mu}^\dagger + a_{\mathbf{k}\mu}^\dagger a_{\mathbf{k}\mu} \right)$$

to the integral of $\mathbf{E}^2 + \mathbf{B}^2$.

Terms Containing Two Annihilation or Two Creation Operators

For the two-annihilation term, the spatial exponential is

$$e^{i(\mathbf{k}+\mathbf{k}')\cdot\mathbf{r}}.$$

Therefore,

$$\int_V d^3r\, e^{i(\mathbf{k}+\mathbf{k}')\cdot\mathbf{r}} = V\delta_{\mathbf{k}',-\mathbf{k}}.$$

Only $\mathbf{k}' = -\mathbf{k}$ survives. For these terms,

$$D_{\mathbf{k}\mu,-\mathbf{k}\mu'} = \hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{-\mathbf{k}\mu'} + \hat{\mathbf{b}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{b}}_{-\mathbf{k}\mu'}.$$

Now,

$$\hat{\mathbf{b}}_{\mathbf{k}\mu} = \hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\mathbf{k}\mu},$$

whereas

$$\hat{\mathbf{b}}_{-\mathbf{k}\mu'} = -\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{-\mathbf{k}\mu'}.$$

Consequently,

$$\begin{aligned} \hat{\mathbf{b}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{b}}_{-\mathbf{k}\mu'} &= -\left(\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\mathbf{k}\mu} \right) \cdot \left(\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{-\mathbf{k}\mu'} \right) \\ &= -\hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{-\mathbf{k}\mu'}. \end{aligned}$$

In the last step, we used the fact that both polarization vectors are perpendicular to $\mathbf{k}$. It follows that

$$D_{\mathbf{k}\mu,-\mathbf{k}\mu'} = 0.$$

Thus, the terms containing two annihilation operators cancel exactly between $\mathbf{E}^2$ and $\mathbf{B}^2$. The same argument shows that the two-creation terms also cancel. This cancellation does not require

any special choice relating the polarization basis at $\mathbf{k}$ to the basis at $-\mathbf{k}$.

Field Energy

Only the terms containing one creation and one annihilation operator remain. Including the factor of $1/2$ in the definition of the energy,

$$\begin{aligned} H_{\text{field}} &= \frac{1}{2}\int_V d^3r\ \left(\mathbf{E}^2 + \mathbf{B}^2\right) \\ &= \frac{1}{2}\left[2V\sum_{\mathbf{k},\mu}\mathcal{E}_{\mathbf{k}}^2\left(a_{\mathbf{k}\mu}a_{\mathbf{k}\mu}^\dagger + a_{\mathbf{k}\mu}^\dagger a_{\mathbf{k}\mu}\right)\right]. \end{aligned}$$

Therefore,

$$\boxed{H_{\text{field}} = V\sum_{\mathbf{k},\mu}\mathcal{E}_{\mathbf{k}}^2\left(a_{\mathbf{k}\mu}a_{\mathbf{k}\mu}^\dagger + a_{\mathbf{k}\mu}^\dagger a_{\mathbf{k}\mu}\right).}$$

References

[1] Robert Bennett, Thomas M. Barlow, and Almut Beige. "A Physically Motivated Quantization of the Electromagnetic Field". In: *European Journal of Physics* 37.1 (2016), p. 014001. DOI: 10.1088/0143-0807/37/1/014001. URL: https://doi.org/10.1088/0143-0807/37/1/014001.